HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Supported the **Permanent Way design team** by performing **data analysis** and executing **design calculations** for critical rail infrastructure projects.
- Assisted in drafting technical specifications, proposals, and detailed investigation reports for **railway engineering** projects, ensuring adherence to best practices.
- Contributed to **cross-disciplinary design coordination**, integrating inter-project interfaces using the latest **3D modelling software** and engineering tools.
- Applied **analytical techniques** to large datasets to identify patterns and inform design decisions for sustainable transport solutions.
- Collaborated with **senior engineers** and project managers to assimilate best practices and contribute creative solutions to complex design challenges.
- Developed **Python scripts** to automate data processing and reporting tasks, improving efficiency for the rail design team by an estimated **15%**.
- Managed **production support** for enterprise applications, resolving P1/P2 incidents and maintaining SLAs within a high-availability environment.

SKILLS

Rail Engineering – Permanent Way Design, Rail Track Design, Railway Engineering, Sustainable Transport

Software & Tools – 3D Modelling Software, CAD Software, Project Management Software, Data Analysis Tools, Python, SQL

Engineering Disciplines – Civil Engineering, Mechanical Engineering, Transportation Engineering, Structural Engineering

Project Delivery – Design Calculations, Technical Specifications, Project Drawings, Interdisciplinary Coordination, Reporting, Investigations

Methodologies – Agile Methodologies, Scrum, Best Practices

PROJECTS

Rail Infrastructure Design & Analysis Support

- Performed **data analysis** on track geometry and load data using **Python** and **SQL** to support permanent way design calculations for a major regional rail expansion project.
- Generated **project drawings** and preliminary **3D models** based on design team specifications, ensuring accurate representation of track layouts and associated infrastructure.
- Assisted in the preparation of technical reports and design documentation, contributing to the successful submission for a **sustainable transport initiative**.

Production System Automation & Monitoring

- Developed **Python scripts** to automate the extraction and aggregation of operational data from production systems, reducing manual reporting effort by approximately **20%**.
- Implemented monitoring dashboards using **Splunk** to track key performance indicators and alert on potential issues within enterprise applications, ensuring high availability.
- Participated in **incident response** for critical systems, performing root cause analysis and implementing corrective actions to maintain service level agreements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)